

Characteristics of Non-chordate Animals

Aim

To study characteristics of non-chordates (1) Earthworm (2) Cockroach.

Requirements

Specimens of earthworm and cockroach.

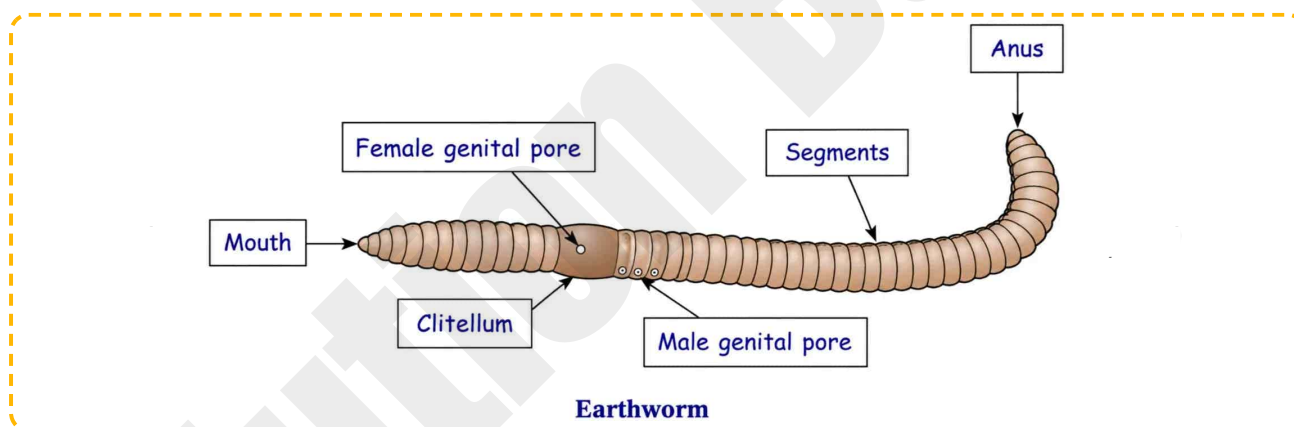
Procedure

Observe the specimens carefully, note down their characteristics, and classify them accordingly.

1. Earthworm

Diagram

(Label the different parts in the given diagram)



Classification of Earthworm:

Kingdom : Animalia

Subkingdom : Non-chordata

Phylum : Annelida

Scientific name : *Pheretima posthuma*

Observations: Characteristics of Earthworm

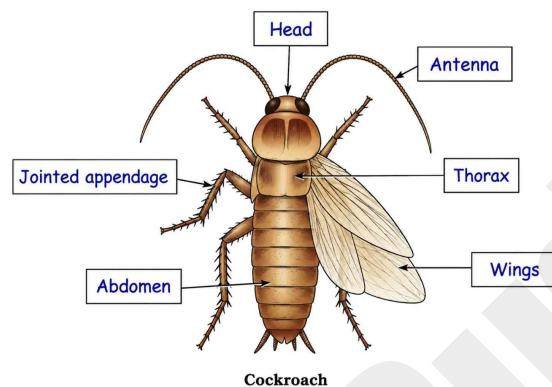
1. The body of earthworm is elongated and cylindrical. It has mouth and anus at the opposite ends.

2. The body is metamerically segmented, with about 100 to 125 segments in the body.
3. A prominent brown-coloured band called clitellum is present in the anterior region of the body covering the segments 14th, 15th and 16th.
4. Earthworm is a bisexual animal crawling in soil. It is said to be a friend of farmer.
5. It is used for vermicompost, therefore they play an important role in solid waste management.

2. Cockroach

Diagram

(Label the different parts in the given diagram)



Classification of Cockroach:

Kingdom : Animalia

Subkingdom : Non-chordata

Phylum : Arthropoda

Class : Insecta

Scientific name : *Periplaneta americana*

Observations: Characteristics of Cockroach

1. The body of cockroach is narrow, bilaterally symmetrical and dorsoventrally flattened.
2. The whole body is covered externally by a chitinous cuticle.
3. The body of cockroach is divisible into three parts, viz., head, thorax and abdomen.
4. The head on its each side bears a large compound eye and a long, slender and segmented antenna.
5. There are 3 pairs of walking legs.
6. Cockroach is a nocturnal pest which is carrier of diseases.

Multiple Choice Questions

1. To which phylum does Earthworm belong?

- (A) Platyhelminthes (B) Aschelminthes
(C) Annelida (D) Arthropoda

Answer: (C) Annelida

2. Cockroach is an insect but spider is not because

- (A) cockroach and spiders belong to different phyla.
(B) cockroach has 3 pairs of legs and spider has 4 pairs of legs.
(C) cockroach has jointed appendages but spider does not have.
(D) spider can build a web but cockroach cannot.

Answer: (B) cockroach has 3 pairs of legs and spider has 4 pairs of legs.

3. Earthworm is called a friend of farmer but cockroach is not. Why?

- (A) Cockroach is a pest and carrier of some human diseases.
(B) Cockroach cannot dig the soil.
(C) Cockroach is hard bodied insect.
(D) Cockroaches do not stay on the fields.

Answer: (A) Cockroach is a pest and carrier of some human diseases.

4. The only common character between earthworm and cockroach is ...

- (A) both breathe by tracheae.
(B) both excrete by nephridia.
(C) both have spiracles.
(D) both are non-chordates.

Answer: (D) both are non-chordates.

5. Clitellum of earthworm is present in which segments?

- (A) 8, 9, 10 (B) 11, 12, 13 (C) 14, 15, 16 (D) 16, 17, 18

Answer: (C) 14, 15, 16

Remember

In Non-chordates, nerve cord is ventral, solid and double.

Date: _____

Teacher's Signature: _____